NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

Rapid Trials Lab mainline

Group Rapid Trials Lab

Status Promoted

Period 2026-10-15 23:46 — 2026-11-07 03:29

Stage conclusion

First choice is K-Dense, with Academic as a style reference; they handle only styling and export: layout must be written by hand and structure checked by a person

Project report

Redrawing the Graph-DeepONet final network architecture diagram with scientific plotting Skills

Objective

Verify the repositories and licenses of 6 candidate Skills (AgentFigureGallery, Nature Skill, Academic Figure Skill, Origin Style Replot, figures4papers, K-Dense Scientific Visualization), select at most 2 (prioritizing those that output editable files) to test-draw the "final network architecture diagram" of the openclaw urban wind-field Graph-DeepONet project, and render a worth-using / drop verdict.

How the test target was selected

Candidates: the 04-09 figure (UNet Speed Distillation Round 34, Global Best; boxes and connector lines overlap), checked against later final versions in the same folder: the 7-level UNet final diagram (04-15) and the project notes (recording the best model as HybridGraphCNN+FineGraph48+Fourier64, with no existing diagram). Which was chosen and why: both the round34 original figure (04-09) and the 7-level UNet final diagram (04-15) belong to a separate UNet reproduction track (`auto_v2`, Lu et al. 2025) and do not depict Graph-DeepONet. The optimal architecture of Graph-DeepONet exists only in text notes (04-02), where lines 35–36 specify a HybridGraphCNN + FineGraph48 + Fourier64 trunk; therefore a schematic was drawn from the text record, with any unrecorded layer counts or channel dimensions explicitly labeled "unrecorded" (original text verified by the lead on 10-16). To answer the question the PI raised ("can the overlaps in the original figure be fixed?"), a supplementary test also redrew the round34 original figure.

Current status

10-16 03:20 Wrap-up: Worth using (with limitations), per the headline conclusion. Graph-DeepONet has no existing final architecture diagram, only text records; the schematic drawn from those records is labeled "not verified against code" and cannot be used as a final journal submission figure.

Security check

See the audit report. Correction by the lead: the `eval(parse(...))` call in Academic's compose script appears only inside a docstring usage example and is never executed anywhere in the repository (verified via `grep`), representing low risk and not grounds for disqualification.

Trials and results

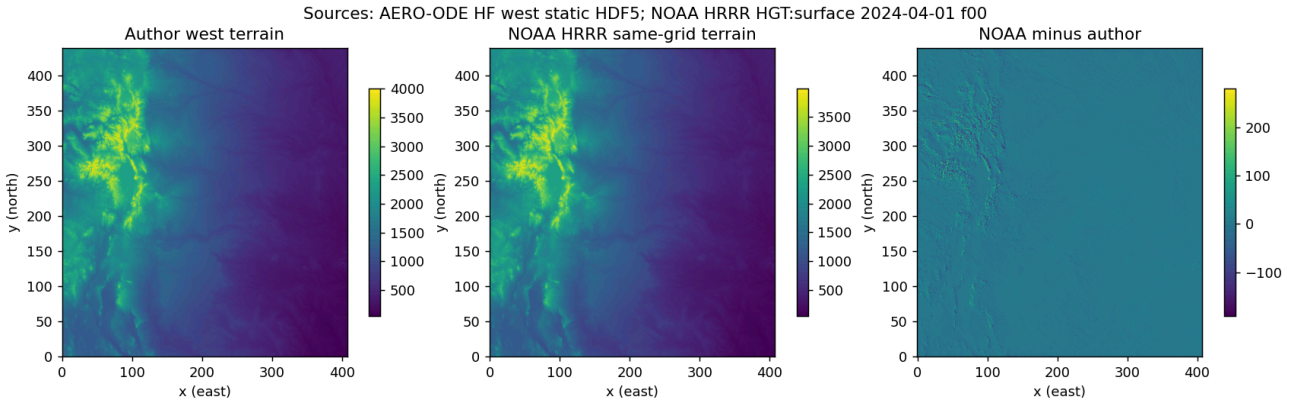
Date	Action taken	One-line result
10-16	Supplementary test: redrew the round34 original figure with both Skills	Verified by the lead: title, subtitle, numbers across all 9 modules, 3 skip connections, and auxiliary head all match the original figure; all box-on-box overlaps and curve-through-box crossings in the original are eliminated; each SVG has 38 editable text elements and the PDFs have embedded fonts
10-16	Audited 6 repositories: 5 located, no matching repository found for Origin Style Replot; figures4papers is CC BY-NC	Selected Academic Figure Skill (Apache-2.0) and K-Dense `scientific-visualization` (MIT)

Date	Action taken	One-line result
10-16	Drew Graph-DeepONet schematic from text records	Both figures are consistent in content, overlap-free, and have editable SVG text; the K-Dense version has excessive outer margins; all unrecorded parameters are explicitly labeled without fabricated numbers

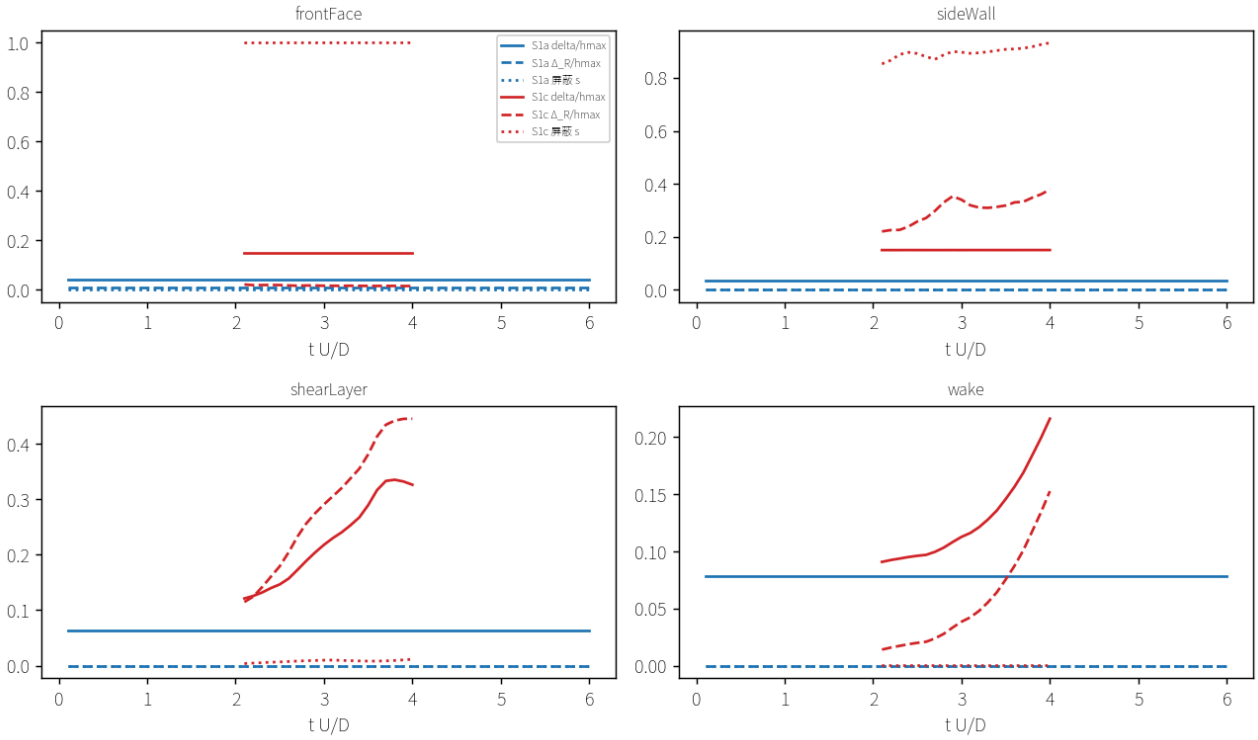
Open items and next steps

- Hand implementation recommendation to Platform Lead: equip the plotting bot (`figures`) only with the `scientific-visualization` subdirectory of K-Dense pinned to commit `a1043ef544`, without installing the full 177-skill repository; from Academic, take only the style reference documents under `references`. Delivery gate: item-by-item checklist against the original figure/notes + visual PNG inspection + editable SVG text.
- If a final submission figure for Graph-DeepONet is needed: first obtain the source code or configuration of the best model to verify the exact layer counts and channel dimensions.

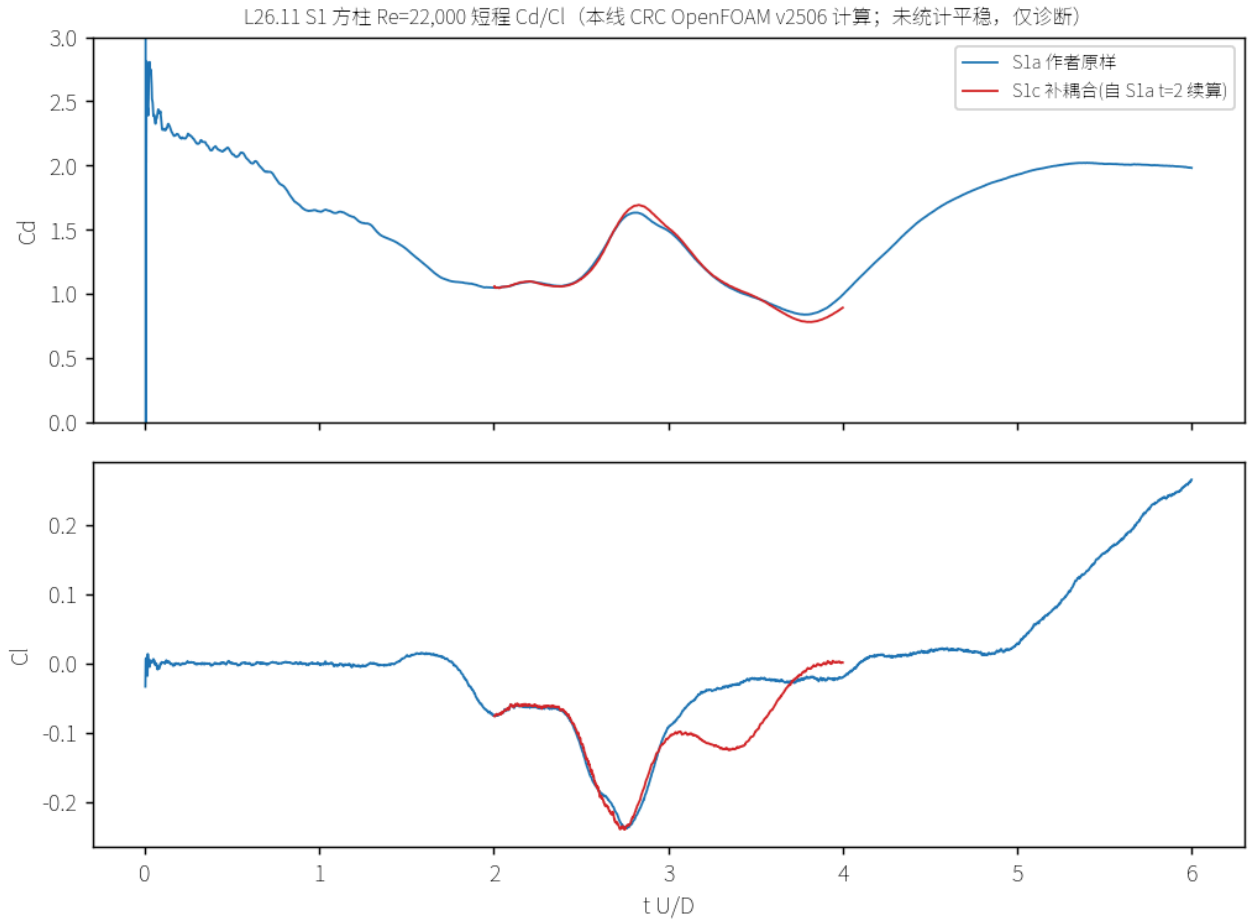
Key figures



terrain alignment diag



S1 zone delta



S1 CdCl

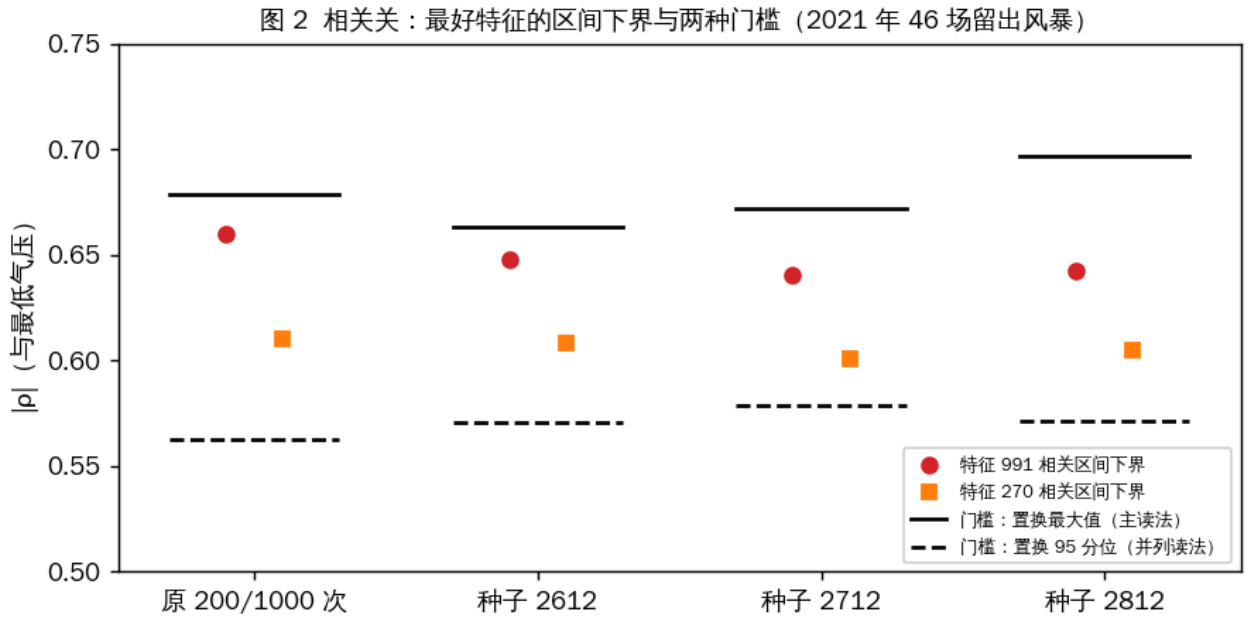


图2 相关门槛

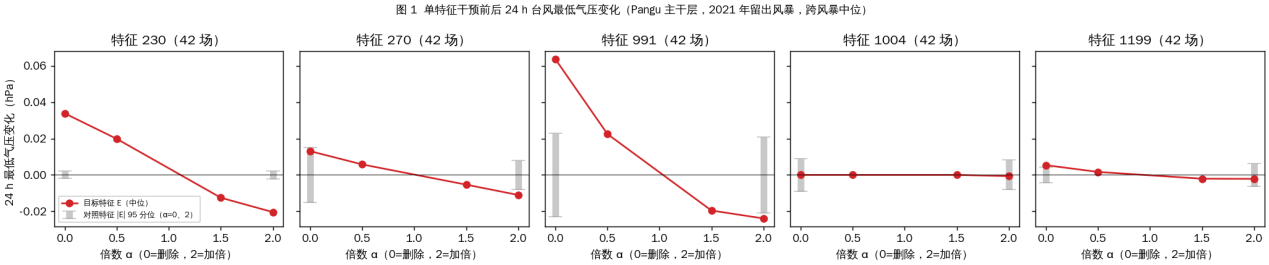

图 1 单特征干预前后 24 h 台风最低气压变化（Pangu 主干层，2021 年留出风暴，跨风暴中位）
特征 230（42 场）
特征 270（42 场）
特征 991（42 场）
特征 1004（42 场）
特征 1199（42 场）
24 h 最低气压变化（hPa）
0.06
0.04
0.02
0.00
-0.02
目标特征 E（中位）
对照特征 |E| 95 分位（α=0，2）
0.0
0.5
1.0
1.5
2.0
倍数 α（0=删除，2=加倍）

图1 干预效应